PETROLEUM AND NATURAL GAS REGULATORY BOARD (PNGRB)
INCIDENT INVESTIGATION REPORT

Document Ref: PNGRB/INV/2025/0412
Date of Incident: 14/03/2025
Location: Hazira-VTP Pipeline Section, District Surat, Gujarat
Operator / Entity Name: GAIL (India) Limited
Pipeline Type: Cross-Country Natural Gas Trunkline (30-inch High Pressure)

1. Executive Summary
On 14th March 2025 at approximately 10:15 AM, a high-pressure natural gas release occurred during hot work n

2. Casualties and Human Impact
Fatalities: 0
Injuries: 2 technician injuries (burns and minor blast injury, hospitalized, non-fatal).
Casualties Detail: Maintenance technicians received thermal burns from localized ignition.

3. Findings / Key Observations
- Hot work (welding and grinding) commenced within 3 meters of pressurized natural gas flange.
- Combustible gas detector was uncalibrated and continuous atmospheric monitoring was not performed.
- Energy isolation tagout (LOTO) valve V-102 was partially passing gas due to worn seal.
- Emergency Shutdown (ESD) valve at upstream terminal took 14 minutes to activate manually.

4. Root Cause Analysis
- Primary Cause: Inadequate Permit-to-Work (PTW) verification and failure to enforce Energy Isolation.
- Underlying Cause: Deficient pre-job safety risk assessment (JSA) for hot work near active pipeline.
- Management Lapses: Ineffective contractor safety supervision during turnarounds.

5. Failed Controls / Safeguards
- Control Failure 1: Mandatory gas testing not completed prior to hot work permit sign-off.
- Control Failure 2: Positive physical blind isolation (LOTO) omitted in favor of single valve closure.
- Control Failure 3: Fire watch personnel lacked active flame suppression equipment at workspace.

6. Financial and Property Damage
Property Damage Cost: INR 45 Lakhs (equipment replacement and line repair)
Gas Loss Quantity: 18,500 SCM natural gas vented during emergency containment.
Environmental Impact: Minimal ground impact; localized gas dispersion.

7. Recommendations / Corrective Actions
- Recommendation 1: Re-certify all PTW issuers and enforce double-block-and-bleed LOTO isolation for hot wor
- Recommendation 2: Mandate calibrated multi-gas detectors with continuous audio-visual alarms at all hot work
- Recommendation 3: Conduct audit of ESD automated actuator closure times across Surat pipeline network.
- Recommendation 4: Implement contractor safety qualification reviews prior to high-risk maintenance permits.

Report Signed By:
Senior Incident Inspector, PNGRB Safety Committee
Date: 28/03/2025